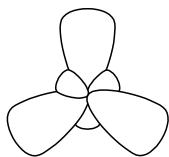


1 (a) (i) Hybridisation:  $sp^2$

[3]



bond angle =  $117^\circ$  (accept  $110^\circ < \text{angle} < 120^\circ$ )

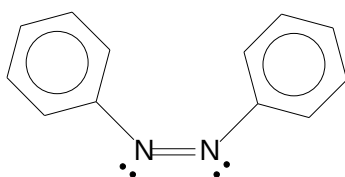
[1]: hybridisation

[1]: hybrid orbitals (allow ecf if hybridisation was wrongly determined)

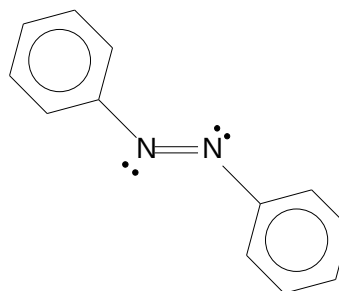
[1]: bond angle (ecf)

(ii) cis-trans isomerism

[1]



cis



trans

[1]: type of isomerism

[1]: structure with correct label (ignore if lp are missing)

(iii) cis-isomer

[2]

For the cis-isomer, bending / flexing occurred which allows the microstructure to peel away from the surface.

(b)

| Observation                                                                           | Type of reaction                | Deduction                                                                                    |
|---------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------|
| Compound <b>U</b> , $C_{17}H_{19}N_3O_3$ is not soluble in $NaOH(aq)$ and $HCl(aq)$ . | (absence of acid-base reaction) | <b>U</b> likely to contain amide.                                                            |
| <b>U</b> gives 2 products when heated under reflux with $Cr_2O_7^{2-}/H^+$ .          | hydrolysis                      | <b>U</b> contains amide or ester group.                                                      |
| Orange $Cr_2O_7^{2-}/H^+$ changes from orange to green during the process.            | oxidation                       | <b>U</b> or its hydrolysed products contain $1^\circ$ / $2^\circ$ alcohol or aldehyde group. |
| <b>V</b> , $C_{13}H_{10}N_2O_2$ has a proton chemical shift of 13.0 ppm.              |                                 | <b>V</b> contains $-COOH$ .<br><b>W</b> contains 4 C atoms and 1 N atom.                     |

[2]

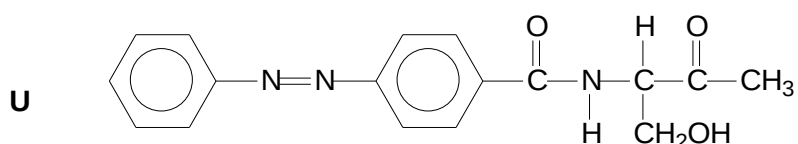
|                                                                                                               |                                     |                                                                     |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------|
| <b>W</b> gives an orange precipitate when treated with 2,4-DNPH but it has no reaction with Tollens' reagent. | condensation / addition–elimination | <b>W</b> contains ketone group.                                     |
| <b>W</b> gives a yellow precipitate and <b>X</b> when warmed with $I_2/NaOH$ .                                | oxidation                           | <b>W</b> contains $RCOCH_3$ .<br><b>X</b> is a sodium salt          |
| Effervescence was observed when <b>W</b> was treated with $NaHCO_3(aq)$ .                                     | acid–base                           | <b>W</b> contains $-COOH$                                           |
| When <b>W</b> is heated with excess $CH_3Cl$ , it gives <b>Y</b> , $C_7H_{14}NO_3Cl$ , as the major product.  | nucleophilic substitution           | <b>W</b> contains $1^\circ$ amine<br><b>Y</b> is a quaternary salt. |

[4]: 14–18 points

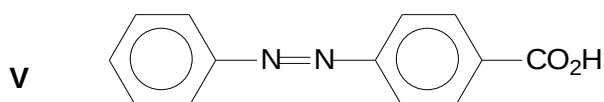
[3]: 10–13 points

[2]: 7–9 points

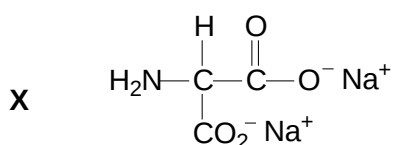
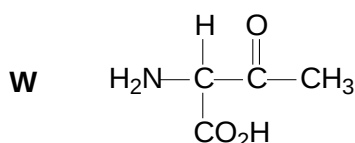
[1]: 3–6 points



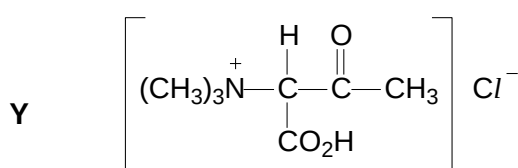
(ecf from **V** and **W**)



(allow  $-COOH$  group to be in other positions)



(ecf from **W**)



(ecf from **W**)

[1] each correct structure

(overall only penalised once if final answers are not given to 1 dp for this question)

2 (a) (i)  $K_{sp} = [Ag^+] [I^-]$  [1]

(ii)  $[Ag^+] = [I^-] = [AgI(aq)]$  [1]

$$K_{sp} = (8.9 \times 10^{-9})^2 \\ = \underline{7.9 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}}$$

[1]: answer given to 1 dp

(iii)  $[Ag^+] = \frac{1}{2} \left( \frac{0.025}{[107.9 + 14.0 + 3(16.0)]} \right)$  [2]  
 $= \underline{7.4 \times 10^{-5} \text{ mol dm}^{-3}}$

Let the solubility of AgI, after addition of AgNO<sub>3</sub>, be s.

$$K_{sp} = [Ag^+] [I^-]$$

$$7.92 \times 10^{-17} = (7.36 \times 10^{-5} + s)(s)$$

Since solubility of AgI in water is  $\ll 10^{-5}$ ,  $(7.36 \times 10^{-5} + s) \approx 7.36 \times 10^{-5}$

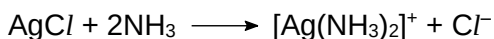
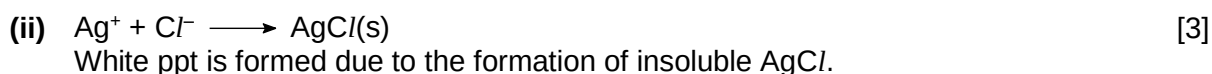
$$s = 1.08 \times 10^{-12} \text{ mol dm}^{-3}$$

$$\text{Mass of AgI ppted} = (8.9 \times 10^{-9} - 1.08 \times 10^{-12}) \times 2 \times (107.9 + 126.9) \\ = \underline{4.2 \times 10^{-6} \text{ g}}$$

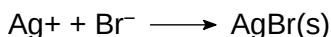
[1]:  $[Ag^+]$  added to 2 dm<sup>3</sup> saturated solution

[1]: mass of AgCl ppted

(b) (i)  $\Delta G_{ppt}^\ominus = 2.303 RT \log_{10} K_{sp}$  [1]  
 $= 2.303(8.31)(298)(\log 2.0 \times 10^{-10}) \text{ J mol}^{-1}$   
 $= \underline{-5.5 \times 10^4 \text{ J mol}^{-1}}$



White ppt dissolves due to the formation of a complex ion.



Cream ppt is obtained when NaBr is added to the resultant mixture because  $K_{sp}$  of AgBr  $< K_{sp}$  of AgCl

[1] each equation and explanation

(c)  $\Delta G_{ppt}^\ominus = 2.303 RT \log_{10} K_{sp}$  [2]  
 $= 2.303(8.31)(298)(\log_{10} 1.006)$   
 $= \underline{+14.8 \text{ J mol}^{-1}}$

Since  $\Delta G_{ppt}^\ominus$  is +ve, precipitation will not occur  
 i.e. AgF is soluble

[1]:  $\Delta G_{ppt}^\ominus$  calculated

[1]: conclusion (ecf)



- 3 (a) (i) *Le Chatelier's Principle* states that if a change is made to a reversible reaction in dynamic equilibrium, the position of equilibrium shifts so as to minimise that change and to re-establish equilibrium. [2]

A high temperature favours the forward endothermic reaction which helps to absorb the excess heat. Position of equilibrium lies to the right for stage 1.

[1]: stating Le Chatelier's Principle

[1]: explanation



(iii)  $\Delta H_f^\ominus = \Sigma \text{BE}(\text{bonds broken}) - \Sigma \text{BE}(\text{bonds formed})$  [2]  
 $283 = 2\text{BE}(\text{C}=\text{O}) - [\text{BE}(\text{C}\equiv\text{O}) + \frac{1}{2}\text{BE}(\text{O}=\text{O})]$   
 $\text{BE}(\text{C}\equiv\text{O}) = 2(805) - 283 - \frac{1}{2}(496)$   
 $= \underline{+1079 \text{ kJ mol}^{-1}}$

[1]: correct identification of bonds broken and bonds formed

[1]:  $\text{BE}(\text{C}\equiv\text{O})$  (ecf if 740 is used as  $\text{BE}(\text{C}=\text{O})$ ; i.e.  $+949 \text{ kJ mol}^{-1}$ )

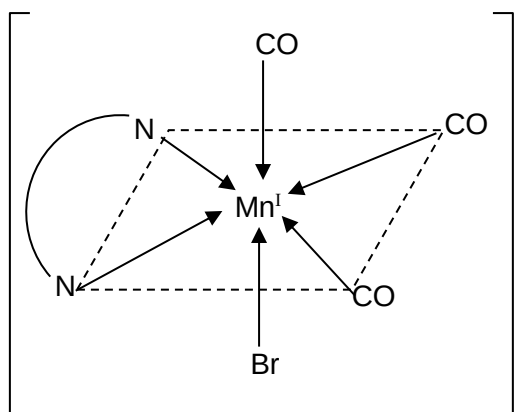
(iv)  $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$  [2]  
 $237 = 283 - (298)(\Delta S^\ominus)$   
 $\Delta S^\ominus = \underline{+0.154 \text{ kJ K}^{-1}\text{mol}^{-1}}$   
 $= \underline{+154 \text{ J K}^{-1}\text{mol}^{-1}}$

Entropy change for the reaction is positive, indicating an increase in disorder of system / ways of distributing energy as seen from the equation in (ii) where there is an increase in number of moles of gaseous particles.

[1]:  $\Delta S^\ominus$ , include units

[1]: explaining link between entropy change and disorder of reaction

- (b) (i) [2]



shape: octahedral

[1]: structure (dative bonds of bidentate ligand cannot be both on axial position)

[1]: shape

- (ii) steps 2 and 6 [1]

- (iii) step 1: reduction [1]  
 step 4: acid base reaction

- (iv) In the presence of ligands, the partially filled 3d orbitals of Mn are split into two levels (non-degenerate) with a small energy gap (d orbital splitting). [3]  
 When light passes through, energy corresponding to the blue light is absorbed and an electron in a lower energy d-orbital is promoted to a higher energy d-orbital. (d-d transition).  
 The orange colour of the solution is the complementary colour of the blue light absorbed.

[1]: partially filled 3d-orbital; small energy gap

[1]: d-d transition

[1]: absorption of blue (complementary to orange)

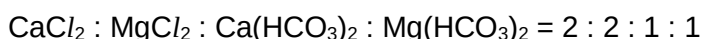
- (c) (i) When pH is high,  $[H^+]$  is low, position of equilibrium shifts to the right and hence more metal carbonate will be precipitated. [1]

(ii)  $[Ca^{2+}] = \frac{0.0120}{40.1} = \underline{2.99 \times 10^{-4} \text{ mol dm}^{-3}}$  [3]

$$[Mg^{2+}] = \frac{0.0073}{24.3} = \underline{3.00 \times 10^{-4} \text{ mol dm}^{-3}}$$

$$[Cl^-] = \frac{0.0284}{35.5} = \underline{8.00 \times 10^{-4} \text{ mol dm}^{-3}}$$

$$[HCO_3^-] = \frac{0.0244}{[1.0 + 12.0 + 3(16.0)]} = \underline{4.00 \times 10^{-4} \text{ mol dm}^{-3}}$$

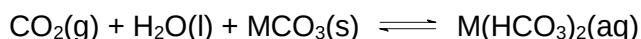


[2]: all concentrations; [1]: any two correct

[1]: correct ratio

- (iii) The rock is made up of calcium carbonate, magnesium carbonate, (calcium chloride and magnesium chloride). [2]

Rainwater containing dissolved atmospheric carbon dioxide shifts the equilibrium position of (2) to the right.



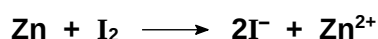
Group 2 bicarbonates are soluble, hence it accounts for the presence of  $HCO_3^-$  in mineral water.

Calcium chloride and magnesium chloride present in the rock also can dissolve in the rainwater, causing  $Ca^{2+}$ ,  $Mg^{2+}$  and  $Cl^-$  to be present in mineral water.

[1]: composition of rock

[1]: explanation using equation (2) to account for presence of bicarbonate

- 4 (a) (i) anode:  $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$  [2]  
 cathode:  $\text{I}_2 + 2\text{e}^- \longrightarrow 2\text{I}^-$



[1]: 2 correct half-equations

[1]: balanced equation

- (ii)  $E_{\text{cell}}^{\ominus} = 0.54 - (-0.76)$  [1]  
 $= +1.30 \text{ V}$

- (iii)  $\Delta G^{\ominus} = -nFE_{\text{cell}}^{\ominus}$  [1]  
 $= -2 \times 96500 \times 1.30$   
 $= -2.50 \times 10^5 \text{ J mol}^{-1}$

(value of n depends on balanced equation in (i))

- (iv) To allow  $\text{Zn}^{2+}$  ions to move from the anode to the cathode during discharging to [1]  
prevent built up of charges / to maintain electrical neutrality.

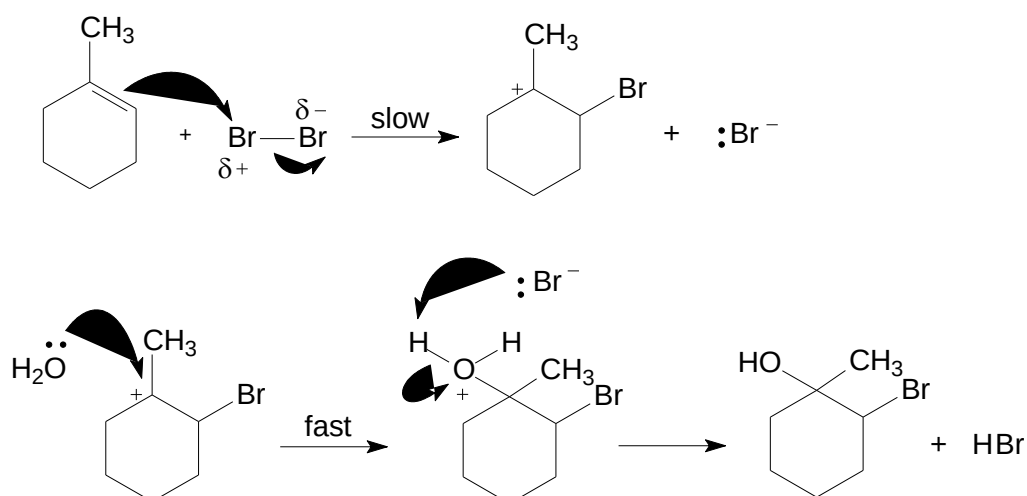


shape: linear

[1]: dot-and-cross

[1]: shape

- (c) (i) electrophilic addition [4]



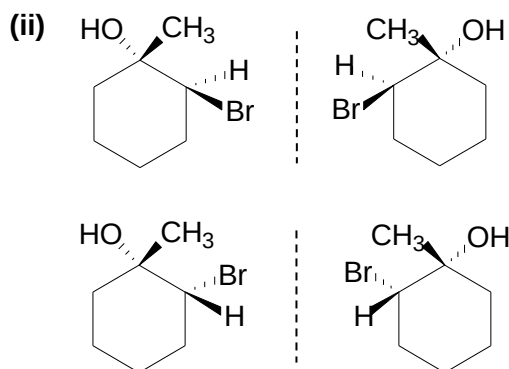
(accept if deprotonation by  $\text{Br}^-$  is shown over 2 steps)

[1]: name of mechanism

[1]: dipoles and charges

[1]: lone pairs and curly arrows

[1]: intermediates and product



[2]

[2]: all 4  
[1]: any 2

- (iii) Reagents and condition: Na metal, room temperature  
Observation: effervescence observed with bromohydrin but no effervescence observed with cyclohexene.

[2]

OR

Reagents and condition: anhydrous  $\text{PCl}_5$  /  $\text{SOCl}_2$ , room temperature

Observation: white fumes observed with bromohydrin but no white fumes observed with cyclohexene.

OR

Reagents and condition:  $\text{KMnO}_4$ , dilute  $\text{H}_2\text{SO}_4$ , cold/heat

Observation: purple  $\text{KMnO}_4$  decolourises for cyclohexene but purple  $\text{KMnO}_4$  remains for bromohydrin.

Alternative (less preferred)

Reagents and condition:  $\text{NaOH(aq)}$  and heat, cool and add  $\text{HNO}_3$  then  $\text{AgNO}_3$ .

Observation: cream ppt observed with bromohydrin but no ppt observed with cyclohexene.

[1]: reagents and condition (do not accept aq.  $\text{Br}_2$ )  
[1]: corresponding observations



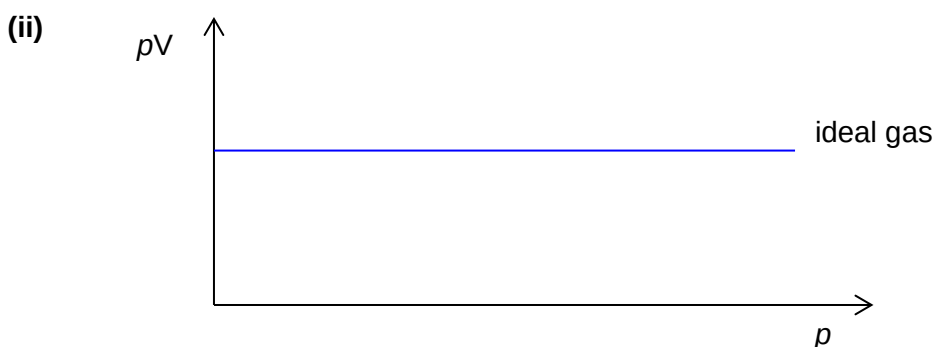
- 5 (a) First I.E. decreases down the group [2]  
nuclear charge increase  
 but electrons are added to next electron/quantum shell  
attraction between the nucleus and the valence electron decreases  
less energy required to remove the valence electron.

[1]: trend

[1]: explanation in terms of increased nuclear charge but weaker attraction due to valence electron being further from nucleus / at a higher energy level

- (b) (i) Gas particles occupies negligible space [2]  
 Negligible forces of attraction between gas particles

[1] each



[1]: sketch complete with ideal gas line and axes labelled

(iii) [3]

| Experiment No. | $pV$ for gas D | $pV$ for gas E |
|----------------|----------------|----------------|
| 1              | 2275           | 2320           |
| 2              | 2198           | 2280           |
| 3              | 2100           | 2190           |
| 4              | 1953           | 2140           |

The  $pV$  values of Gas D deviates more from a constant value than Gas E.  
 Hence, Gas D behaves less like an ideal gas than Gas E.

Gas E is  $H_2$ , which is smaller in size and also has weaker instantaneous dipole-induced dipole interactions between molecules and deviate less from ideality while gas D is  $O_2$  which is a larger molecule with relatively stronger instantaneous dipole-induced dipole interactions.

[1]: all  $pV$  values

[1]: identify gases D and E

[1]: explanation

- (c) (i) Lewis base is defined as an electron pair donor. [1]
- (ii)  $PCl_5 + SO_2(OH)_2 \longrightarrow ClSO_3H + POCl_3 + HCl$  [1]
- (iii)  $H_2SO_4 + H_2O \rightleftharpoons HSO_4^- + H_3O^+$  [1]  
 $ClSO_3H + H_2O \rightleftharpoons ClSO_3^- + H_3O^+$

The  $HSO_4^-$  conjugate base is more stable because there will be greater charge dispersal as it has an additional and more electronegative O. Thus the P.O.E. of  $H_2SO_4$  acid ionisation lie more to the right.

(d) (i) The lone pair of electrons on N is delocalised to the carbonyl carbon and hence it is not available in accepting a proton, making **A** a neutral compound. [1]

(ii) Boil/heat with NaOH(aq), followed by adding Br<sub>2</sub>(aq) to the resultant hydrolysed products. [2]

The product of **A** (phenylamine) will decolourise the orange Br<sub>2</sub>(aq) together with a white ppt but not the other isomer.

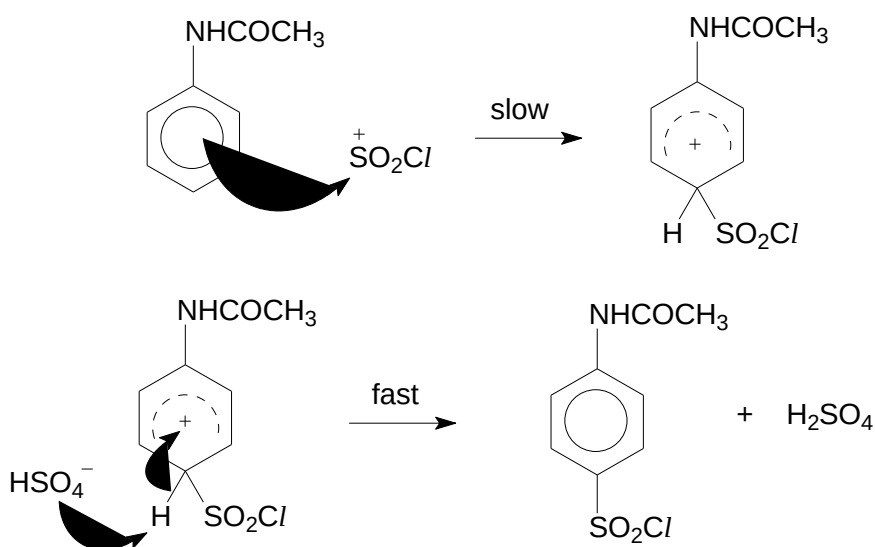
[1]: reagents and condition (accept HCl, if followed by excess NaOH)

[1]: corresponding observations

(iii)  $\text{C}/\text{SO}_3\text{H} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{C}/\text{SO}_2^+ + 2\text{HSO}_4^- + \text{H}_3\text{O}^+$  [1]

(accept  $\text{C}/\text{SO}_3\text{H} + \text{H}_2\text{SO}_4 \longrightarrow \text{C}/\text{SO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$ )

(iv) electrophilic substitution [3]



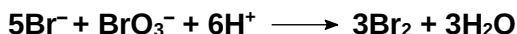
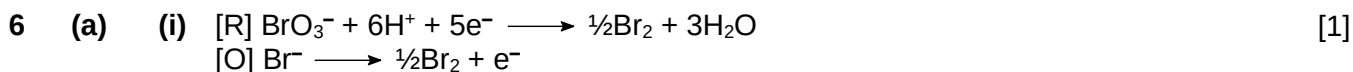
[1]: name of mechanism

[1]: dipoles, charges and lone pairs

[1]: curly arrows, intermediate and regeneration of catalyst

(v) nucleophilic substitution [1]

(vi)  $\text{H}_2\text{O}$  nucleophile competes with  $\text{NH}_3$  nucleophile for the electron deficient S to displace  $\text{Cl}^-$ , affecting the yield of **C**. [1]



(ii) Comparing Expt 1 & 2, [2]

when  $[\text{Br}^-]$  increases by 2 times, rate increases by 2 times  $\left(\frac{8 \times 10^{-2}}{4 \times 10^{-2}}\right)$ .

$\Rightarrow$  rate is directly proportional to  $[\text{Br}^-]$

$\Rightarrow$  reaction is 1<sup>st</sup> order with respect to  $\text{Br}^-$

Comparing Expt 2 & 3,

when  $[\text{BrO}_3^-]$  increases by 2 times, rate increases by 2 times  $\left(\frac{4 \times 10^{-2}}{2 \times 10^{-2}}\right)$ .

$\Rightarrow$  rate is directly proportional to  $[\text{BrO}_3^-]$

$\Rightarrow$  reaction is 1<sup>st</sup> order with respect to  $\text{BrO}_3^-$

$[\text{H}^+]$  at pH 1.00 =  $0.100 \text{ mol dm}^{-3}$

$[\text{H}^+]$  at pH 1.30 =  $0.0501 \text{ mol dm}^{-3}$

Comparing Expt 1 & 4 and let order of reaction with respect to  $\text{H}^+$  be x

$$\frac{8 \times 10^{-2}}{1 \times 10^{-2}} = \frac{(0.10)(0.10)(0.1)^x}{(0.05)(0.10)(0.05)^x}$$

$$x = 2$$

$\Rightarrow$  reaction is 2<sup>nd</sup> order with respect to  $\text{H}^+$

[1] each

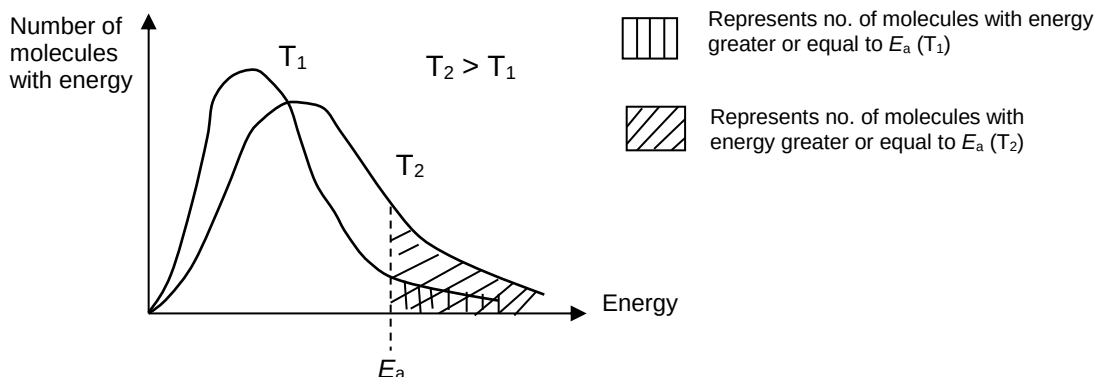
(iii) **rate =  $k [\text{BrO}_3^-] [\text{Br}^-] [\text{H}^+]^2$**  [1]

(iv) Using values from Expt 1, [1]

$$8 \times 10^{-2} = k (0.10) (0.10) (0.10)^2$$

$$k = \underline{800 \text{ mol}^{-3} \text{ dm}^9 \text{ s}^{-1}}$$

(b) [2]



When temperature increases, the (average) kinetic energies of the reactant molecules increase.

As shown on the diagram, the number of reactant molecules with energy greater or equal to the activation energy will increase.

This results in an increase in the frequency of effective collisions. Hence, the rate of reaction increases.

[1]: correct axes, shape of graph (including beginning from origin) and correct shift of graph and shading when T increase

[1]: correct explanation (in terms of ave KE increase and frequency of effective collision increase)

[1]: correct conclusion

- (c) (i) As seen in the equation,  $\ln k = \ln A - \frac{E_a}{R} \left( \frac{1}{T} \right)$  [1]

$$\text{gradient} = -\frac{E_a}{R}$$

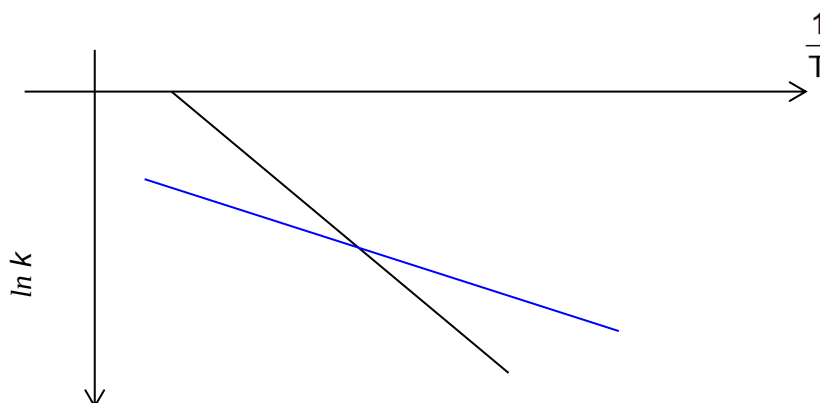
$$\text{Taking points from the graph, gradient} = \frac{-5.20 - (-6.90)}{(3.10 - 3.30) \times 10^{-3}} = -8500$$

$$-8500 = -\frac{E_a}{R}$$

$$E_a = +70600 \text{ J mol}^{-1}$$

- (ii) Both  $\text{BrO}_3^-$  and  $\text{Br}^-$  are negatively charged and they experience inter-electronic repulsion and hence  $E_a$  of the reaction is high. [1]

- (iii) [1]



Since the reaction will have a lower activation energy in the presence of a catalyst, the slope will be less steep.

- (d) (i) [7]

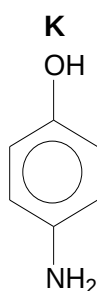
| Observation                                                                                                                        | Type of reaction                                | Deduction                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>K</b> , $\text{C}_6\text{H}_7\text{ON}$ , is not very soluble in water, but dissolves in $\text{HCl(aq)}$ .                     | acid–base                                       | <b>K</b> contains basic <u>amine</u> group.                                                    |
| <b>K</b> also dissolves in $\text{NaOH(aq)}$ , but not in $\text{Na}_2\text{CO}_3\text{(aq)}$ .                                    | acid–base                                       | <b>K</b> contains acidic <u>phenol</u> group<br>Absence of $-\text{CO}_2\text{H}$ .            |
| On reaction with 1 mol of $\text{CH}_3\text{COCl}$ , <b>K</b> forms compound <b>L</b> , $\text{C}_8\text{H}_9\text{O}_2\text{N}$ . | nucleophilic (acyl) substitution / condensation | <b>L</b> contains an <u>amide</u> group or <u>ester</u> group.                                 |
| On reaction with $\text{Br}_2\text{(aq)}$ , <b>L</b> produces compound <b>M</b> , $\text{C}_8\text{H}_7\text{O}_2\text{NBr}_2$ .   | electrophilic substitution                      | 2 x Br substituted suggests that <u>one of the 2, 4 or 6 position</u> of <b>L</b> is occupied. |

|                                                                                                                |                                                 |                                                                                 |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------|
| When <b>K</b> is reacted with 2 mol of ethanoyl chloride, it produces compound <b>N</b> , $C_{10}H_{11}O_3N$ . | nucleophilic (acyl) substitution / condensation | <b>N</b> contains an <u>amide</u> group and an <u>ester</u> group.              |
| <b>N</b> is not soluble in either $HCl(aq)$ or $NaOH(aq)$ .                                                    | –                                               | <u>absence of basic amine and acidic phenol</u> group                           |
| <b>K</b> can be synthesised by treating phenol with dil. $HNO_3$ .                                             | electrophilic substitution                      | <u>4-nitrophenol</u> (or 2-nitrophenol) is formed.                              |
| Followed by reaction with Sn metal and $HCl$ .                                                                 | reduction                                       | 4-nitrophenol is reduced to form <b>K</b> which contains an <u>amine</u> group. |

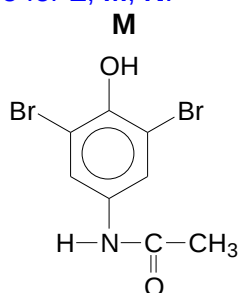
[3]: 14–18 points

[2]: 9–13 points

[1]: 4–8 points

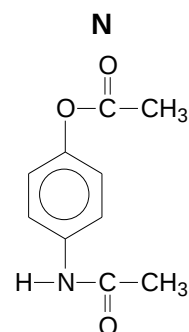
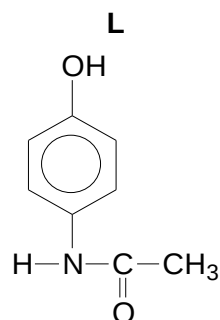


Accepts 2-aminophenol.  
Likewise for **L**, **M**, **N**.

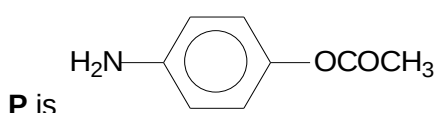


(ecf from **L**, i.e. give **P** instead)

[1] each correct structure



(ii)



[2]

Presence of nucleophilic  $-NH_2$  and  $-OH$  groups in **K** allows the 1 mol of  $CH_3COCl$  to react with the more nucleophilic  $-NH_2$  group to form **L** or to react with the less nucleophilic  $-OH$  group to form minor product **P**.

[1]: structure of **P**

[1]: explanation why there 2 possible products and which is major/minor